

Abstract

Background: Cancer survivorship is rising in the U.S., increasing demand for home health care (HHC). **Objective:** Characterize prevalence, intensity, and payer mix of HHC among adult cancer survivors using MEPS 2016–2019. **Methods:** Survey-weighted analyses; pooled design with annualized weights and year-specific strata/PSU interactions. Logistic (any use), quasi-Poisson (days among users), and log-linear (expenditures among users with >\$0). **Results:** ~One in six survivors used HHC; use and intensity concentrated among socioeconomically vulnerable groups. Financing predominantly public (Medicare/Medicaid), with year-to-year shifts. **Conclusions:** Structural and insurance factors strongly shape HHC utilization; policies targeting high-need survivors may reduce inequities.

Packages

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr   1.5.1
## ✓ ggplot2    3.5.2      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.4
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)
library(survey)
```

```
## Loading required package: grid
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
##
## Loading required package: survival
##
## Attaching package: 'survey'
##
## The following object is masked from 'package:graphics':
##
##   dotchart
```

```
library(broom)
library(srvyr)
```

```
##
## Attaching package: 'srvyr'
##
## The following object is masked from 'package:stats':
##
##   filter
```

Note: We rely on the Full-Year Consolidated (FYC) files, which already contain person-level home health variables and **payer-specific totals**. This avoids double counting from event-level joins.

Helper functions

```
# Read a zipped MEPS file (SSP or DTA) and return a data frame
read_meps_zip <- function(url){
  tf <- tempfile(); download.file(url, tf, mode = "wb")
  files <- unzip(tf, exdir = tempdir())
  ssp <- files[grepl("\\.ssp$", files, ignore.case = TRUE)]
  dta <- files[grepl("\\.dta$", files, ignore.case = TRUE)]
  if(length(ssp)) return(haven::read_xpt(ssp[1]))
  if(length(dta)) return(haven::read_dta(dta[1]))
  stop("No .ssp or .dta found in zip: ", url)
```

```

}

# Utility: detect any available non-melanoma skin cancer flags; returns logical vector
.detect_nmsc <- function(df){
  # Try a few likely variable names; if none exist, return all FALSE
  cand <- intersect(c("CASKINNM", "CANNSKIN", "CASKIN", "CASKINNM16", "CASKIN17", "CASKIN18", "CASKIN19"), names(df))
  if(length(cand)==0) return(rep(FALSE, nrow(df)))
  mat <- do.call(cbind, lapply(cand, function(nm) as.integer(coalesce(df[[nm]], 0)==1)))
  rowSums(mat, na.rm = TRUE) > 0
}

# Define cancer survivor flag (excludes non-melanomatous skin cancer when available)
.define_cancer <- function(df){
  site_vars <- intersect(c("CABLADDR", "CABREAST", "CACERVIX", "CACOLON", "CALUNG", "CALYMPH",
    "CAMELANO", "CAOTHER", "CAPROSTA", "CAUTERUS"), names(df))
  any_site <- if(length(site_vars)){
    rowSums(do.call(cbind, lapply(site_vars, function(nm) as.integer(coalesce(df[[nm]], 0)==1))), na.rm=TRUE) > 0
  } else rep(TRUE, nrow(df))
  nmsc <- .detect_nmsc(df)
  as.integer(coalesce(df$CANCERDX==1, FALSE) & any_site & !nmsc)
}

# Build a per-year FYC loader that harmonizes names we need
load_fyc_year <- function(year){
  yy <- substr(year, 3, 4)
  url <- switch(as.character(year),
    "2016" = "https://meps.ahrq.gov/mepsweb/data_files/pufs/h192ssp.zip",
    "2017" = "https://meps.ahrq.gov/mepsweb/data_files/pufs/h201/h201dta.zip",
    "2018" = "https://meps.ahrq.gov/mepsweb/data_files/pufs/h209/h209dta.zip",
    "2019" = "https://meps.ahrq.gov/mepsweb/data_files/pufs/h216/h216dta.zip",
    stop("Year not supported: ", year)
  )
  fy <- read_meps_zip(url)

  # Coalesce INSCOVxx / INSCOPxx (some years label differs)
  inscov_nm <- paste0("INSCOV", yy)
  inscop_nm <- paste0("INSCOP", yy)
  if(!(inscov_nm %in% names(fy)) && inscop_nm %in% names(fy)){
    fy[[inscov_nm]] <- fy[[inscop_nm]]
  }

  out <- fy %>%
    transmute(
      DUPERSID,
      VARPSU, VARSTR,
      PERWT = .data[[paste0("PERWT", yy, "F")]],
      year = year,
      AGE = .data[[paste0("AGE", yy, "X")]],
      SEX,
      RACE = RACEV1X,
      INSCOV = .data[[inscov_nm]],
      POVCAT = .data[[paste0("POVCAT", yy)]],
      HIDEG,
      MARRY = .data[[paste0("MARRY", yy, "X")]],
      # Home health days (total & by type) and expenditures by payer
      HHTOTD = .data[[paste0("HHTOTD", yy)]],
      HHAGD = .data[[paste0("HHAGD", yy)]],
      HHINDD = .data[[paste0("HHINDD", yy)]],
      HHINFDD = .data[[paste0("HHINFDD", yy)]],
      HHAEXP = .data[[paste0("HHAEXP", yy)]],
      HHNEXP = .data[[paste0("HHNEXP", yy)]],
      HHASLF = .data[[paste0("HHASLF", yy)]],
      HHNSLF = .data[[paste0("HHNSLF", yy)]],
      HHAMCR = .data[[paste0("HHAMCR", yy)]],
      HHNMCR = .data[[paste0("HHNMCR", yy)]],
      HHAMCD = .data[[paste0("HHAMCD", yy)]],
      HHNMCD = .data[[paste0("HHNMCD", yy)]],
      HHAPRV = .data[[paste0("HHAPRV", yy)]],
      HHNPRV = .data[[paste0("HHNPRV", yy)]],
      HHATCH = .data[[paste0("HHATCH", yy)]],
      HHNTCH = .data[[paste0("HHNTCH", yy)]],
      # Cancer flags
      CANCERDX,
      CABLADDR, CABREAST, CACERVIX, CACOLON, CALUNG, CALYMPH,
      CAMELANO, CAOTHER, CAPROSTA, CAUTERUS,
      # If present, keep any skin-related cancer flags to enable exclusion
      !!!dplyr::select(fy, tidyselect::matches("^CA.*SKIN", ignore.case = TRUE))
    ) %>%
    mutate(

```

```

cancerpt = .define_cancer(pick(dplyr::everything())),
homecare_any = as.integer(replace_na(HHTOTD, 0) > 0),
homecare_days = pmax(replace_na(HHTOTD, 0), 0),
exp_total = coalesce(HHAEXP,0) + coalesce(HHNEXP,0),
exp_oop = coalesce(HHASLF,0) + coalesce(HHNSLF,0),
exp_mcr = coalesce(HHAMCR,0) + coalesce(HHNMCR,0),
exp_mcd = coalesce(HHAMCD,0) + coalesce(HHNMCD,0),
exp_prv = coalesce(HHAPRV,0) + coalesce(HHNPRV,0)
)
return(out)
}

# Recode to human-readable factors and drop inapplicable codes
recode_meps <- function(df){
  df %>%
    filter(!is.na(AGE), AGE >= 18) %>%
    mutate(
      SEX = factor(SEX, levels = c(1,2), labels = c("Male","Female")),
      RACE = factor(RACE, levels = c(1,2,3,4,6),
        labels = c("White","Black","Native","AsianPI","Multiple")),
      INSCOV = factor(INSCOV, levels = c(1,2,3),
        labels = c("Private","PublicOnly","Uninsured")),
      POVCAT = factor(POVCAT, levels = c(1,2,3,4,5),
        labels = c("Poor","NearPoor","LowIncome","Middle","High")),
      HIDEG = na_if(HIDEG, -9) |> na_if(-8) |> na_if(-7),
      MARRY = na_if(MARRY, -9) |> na_if(-8) |> na_if(-7) |> na_if(6),
      HIDEG = factor(HIDEG, levels = c(1,2,3,4,5,6,7,8),
        labels = c("NoHS","GED","HS","SomeCollege","AA","BA","MA","PhD")),
      MARRY = factor(MARRY, levels = c(1,2,3,4,5),
        labels = c("Married","Widowed","Divorced","Separated","NeverMarried"))
    )
}

```

Load & harmonize MEPS 2016–2019

```

meps2016 <- load_fyc_year(2016)
meps2017 <- load_fyc_year(2017)
meps2018 <- load_fyc_year(2018)
meps2019 <- load_fyc_year(2019)

# Combine & recode
meps_all <- bind_rows(meps2016, meps2017, meps2018, meps2019) %>%
  recode_meps() %>%
  mutate(
    # Annualize weights for 4-year pooling
    PERWT_adj = PERWT/4,
    # Make strata/PSU unique across years (avoids collisions)
    STRATA_Pooled = interaction(VARSTR, year, drop = TRUE),
    PSU_Pooled = interaction(VARPSU, year, drop = TRUE)
  )

# Survey designs
options(survey.lonely.psu = "adjust")

# Pooled (primary)
meps_design <- svydesign(
  ids = ~PSU_Pooled,
  strata = ~STRATA_Pooled,
  weights= ~PERWT_adj,
  data = meps_all,
  nest = TRUE
)

# Per-year designs (for year-specific estimates/plots)
make_year_design <- function(df, year){
  d <- df %>% filter(year == !!year)
  svydesign(ids = ~VARPSU, strata = ~VARSTR,
    weights = ~PERWT, data = d, nest = TRUE)
}
meps2016_design <- make_year_design(meps_all, 2016)
meps2017_design <- make_year_design(meps_all, 2017)
meps2018_design <- make_year_design(meps_all, 2018)
meps2019_design <- make_year_design(meps_all, 2019)

# Cancer survivors only
cancer_design <- subset(meps_design, cancerpt == 1)
users_design <- subset(cancer_design, homecare_any == 1)
# All HHC users (for cancer vs non-cancer comparisons in expenditure components)
users_design_all <- subset(meps_design, homecare_any == 1)

```

Descriptive results

Prevalence of any HHC (adult cancer survivors)

```
svymean(~homecare_any, cancer_design)
```

```
##              mean      SE
## homecare_any 0.097564 0.0043
```

```
# By sociodemographic characteristics
svyby(~homecare_any, ~SEX, cancer_design, svymean)
```

```
##          SEX homecare_any      se
## Male    Male  0.08835001 0.005952968
## Female  Female  0.10371376 0.005622420
```

```
svyby(~homecare_any, ~RACE, cancer_design, svymean)
```

```
##          RACE homecare_any      se
## White    White  0.09567168 0.004755219
## Black    Black  0.12327460 0.013445277
## Native    Native  0.07726024 0.032562442
## AsianPI  AsianPI  0.04751751 0.016447890
## Multiple Multiple 0.13450123 0.030031669
```

```
svyby(~homecare_any, ~INSCOV, cancer_design, svymean)
```

```
##                INSCOV homecare_any          se
## Private      Private  0.07329546 0.004966007
## PublicOnly PublicOnly 0.14363560 0.007949922
## Uninsured    Uninsured 0.02185803 0.013612753
```

```
svyby(~homecare_any, ~POVCAT, cancer_design, svymean)
```

```
##                POVCAT homecare_any          se
## Poor          Poor    0.17850891 0.015663907
## NearPoor      NearPoor 0.16904257 0.022169100
## LowIncome     LowIncome 0.16347320 0.015838219
## Middle        Middle   0.10226379 0.008644407
## High          High     0.05080947 0.004835989
```

```
svyby(~homecare_any, ~HIDEG, cancer_design, svymean)
```

```
##                HIDEG homecare_any          se
## NoHS          NoHS     0.20776553 0.01557580
## GED           GED      0.11731218 0.02199369
## HS            HS       0.09794341 0.00652071
## SomeCollege   SomeCollege 0.08155879 0.01051077
## AA            AA       0.05130087 0.01009914
## BA            BA       0.05512888 0.01625118
## MA            MA       0.06828019 0.01055392
```

```
svyby(~homecare_any, ~MARRY, cancer_design, svymean)
```

```
##                MARRY homecare_any          se
## Married       Married  0.06610116 0.005249663
## Widowed       Widowed  0.19881326 0.014216825
## Divorced      Divorced 0.10168532 0.009785367
## Separated     Separated 0.06532610 0.020468333
## NeverMarried  NeverMarried 0.11182331 0.014882395
```

Intensity among users (days)

```
svymean(~HHTOTD, users_design)
```

```
##          mean      SE
## HHTOTD 82.022 6.7337
```

```
svymean(~HHAGD + HHINDD + HHINFD, users_design)
```

```
##          mean      SE
## HHAGD 68.8022 6.5858
## HHINDD 3.2806 0.6652
## HHINFD 9.9390 1.5378
```

```
svyby(~HHTOTD, ~SEX, users_design, svymean)
```

```
##          SEX  HHTOTD          se
## Male      Male 80.85952 9.202819
## Female    Female 82.68243 8.221164
```

```
svyby(~HHTOTD, ~RACE, users_design, svymean)
```

```
##          RACE  HHTOTD          se
## White      White 78.56410 7.70162
## Black      Black 105.89866 14.48750
## Native     Native 94.87656 56.54967
## AsianPI    AsianPI 27.39816 14.43114
## Multiple   Multiple 112.51674 30.49834
```

```
svyby(~HHTOTD, ~INSCOV, users_design, svymean)
```

```
##              INSCOV    HHTOTD      se
## Private      Private  52.92850  7.045369
## PublicOnly  PublicOnly 108.36152 10.370743
## Uninsured   Uninsured  18.19011  4.735725
```

```
svyby(~HHTOTD, ~POVCAT, users_design, svymean)
```

```
##              POVCAT    HHTOTD      se
## Poor          Poor  117.67834 13.575212
## NearPoor     NearPoor 132.76249 23.895025
## LowIncome    LowIncome  87.13559 19.190699
## Middle       Middle   60.89074 11.541276
## High         High    55.69955  9.010476
```

```
svyby(~HHTOTD, ~HIDEG, users_design, svymean)
```

```
##              HIDEG    HHTOTD      se
## NoHS          NoHS  134.51448 17.135827
## GED           GED   65.68518 15.128195
## HS            HS    65.19409  6.014702
## SomeCollege  SomeCollege 75.21315 17.505715
## AA           AA    63.26650 20.261561
## BA           BA    16.10839  5.373288
## MA           MA    73.82852 34.103328
```

```
svyby(~HHTOTD, ~MARRY, users_design, svymean)
```

```
##              MARRY    HHTOTD      se
## Married      Married  73.87537 14.273274
## Widowed      Widowed  88.24968  7.273150
## Divorced     Divorced  65.49802  7.969455
## Separated    Separated  56.39561 29.536454
## NeverMarried NeverMarried 124.31754 26.937994
```

Expenditures by payer (users)

```
# Total and by payer among users
svymean(~(HHAEXP + HHNEXP), users_design)
```

```
##          mean      SE
## HHAEXP 9196.18 1573.916
## HHNEXP  166.79   39.091
```

```
svymean(~(HHASLF + HHNSLF), users_design) # OOP
```

```
##          mean      SE
## HHASLF 106.637 40.047
## HHNSLF  94.688 25.753
```

```
svymean(~(HHAMCR + HHNMCR), users_design) # Medicare
```

```
##          mean      SE
## HHAMCR 5128.7421 1459.4041
## HHNMCR   1.6541   1.1275
```

```
svymean(~(HHAMCD + HHNMCD), users_design) # Medicaid
```

```
##          mean      SE
## HHAMCD 2967.100 634.735
## HHNMCD  48.398 25.743
```

```
svymean(~(HHAPRV + HHNPRV), users_design) # Private
```

##	mean	SE
## HHAPRV	254.336	81.0272
## HHNPRV	2.443	1.6859

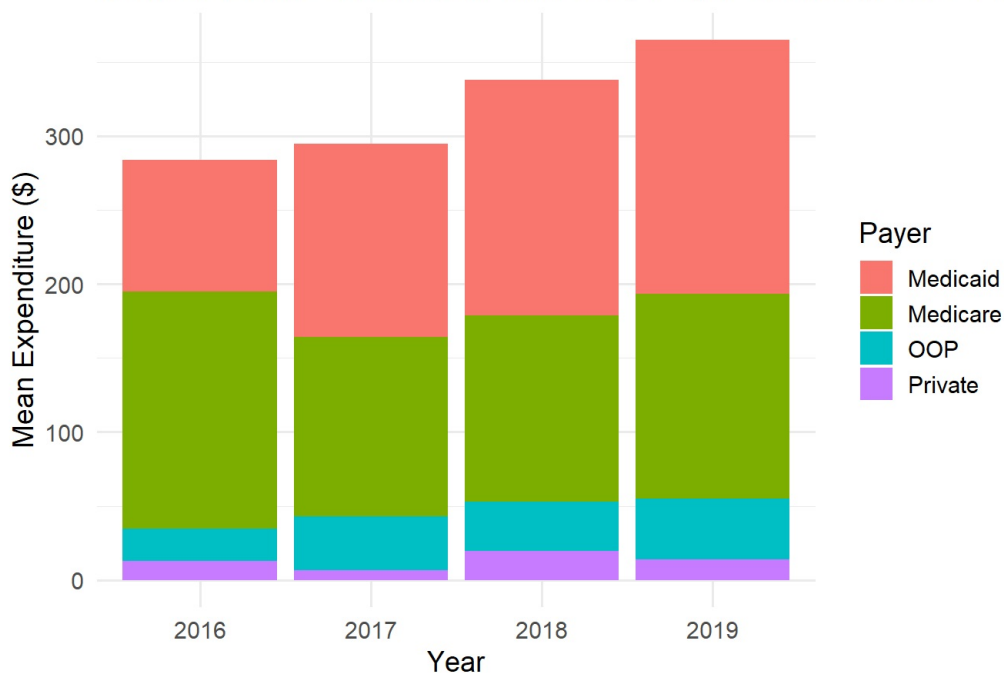
Year-specific payer means (stacked bars)

```
get_payer_means <- function(design, year){
  tibble(
    year = year,
    Medicare = coef(svymean(~I(HHAMCR + HHNMCR), design)),
    Medicaid = coef(svymean(~I(HHAMCD + HHNMCD), design)),
    Private = coef(svymean(~I(HHAPRV + HHNPRV), design)),
    OOP = coef(svymean(~I(HHASLF + HHNSLF), design))
  ) %>%
  pivot_longer(-year, names_to = "source", values_to = "exp")
}

by_year <- bind_rows(
  get_payer_means(meps2016_design, 2016),
  get_payer_means(meps2017_design, 2017),
  get_payer_means(meps2018_design, 2018),
  get_payer_means(meps2019_design, 2019)
)

by_year %>%
  ggplot(aes(x = factor(year), y = exp, fill = source)) +
  geom_col() +
  labs(x = "Year", y = "Mean Expenditure ($)",
       title = "Home Health Care Expenditures by Payer Source, 2016–2019",
       fill = "Payer") +
  theme_minimal(base_size = 14)
```

Home Health Care Expenditures by Payer Source, 2016–2019



Expanded HHC components: Agency vs Non-agency & Charges (Cancer vs Non-cancer)

```
# Ensure charges_total is in the survey design
meps_design <- update(meps_design,
  charges_total = ifelse(is.na(HHATCH), 0, HHATCH) + ifelse(is.na(HHNTCH), 0, HHNTCH)
)

# Refresh subsets so they inherit the new variable
users_design_all <- subset(meps_design, homecare_any == 1)
cancer_design <- subset(meps_design, cancerpt == 1)
users_design <- subset(cancer_design, homecare_any == 1)

# POOLED (2016-2019): means among HHC users by cancer status
exp_pooled_cn_components <- svyby(
  ~HHAEXP + HHNEXP + exp_total + charges_total +
    HHAMCR + HHNMCR + HHAMCD + HHNMCD + HHAPRV + HHNPRV + HHASLF + HHNSLF,
  ~factor(cancerpt, labels=c("Non-cancer", "Cancer")),
  design = users_design_all, svymean, na.rm = TRUE)
exp_pooled_cn_components <- svyby(
  ~HHAEXP + HHNEXP + exp_total + charges_total +
    HHAMCR + HHNMCR + HHAMCD + HHNMCD + HHAPRV + HHNPRV + HHASLF + HHNSLF,
  ~factor(cancerpt, labels=c("Non-cancer", "Cancer")),
  design = users_design_all, svymean, na.rm = TRUE)
exp_pooled_cn_components
```

```
##          factor(cancerpt, labels = c("Non-cancer", "Cancer"))    HHAEXP
## Non-cancer                                                    Non-cancer 7569.464
## Cancer                                                         Cancer 9196.177
##          HHNEXP exp_total charges_total    HHAMCR    HHNMCR    HHAMCD
## Non-cancer 769.4196 8338.883    10191.85 2870.039 19.027668 3487.184
## Cancer    166.7921 9362.969    10839.14 5128.742  1.654075 2967.100
##          HHNMCD HHAPRV    HHNPRV    HHASLF    HHNSLF se.HHAEXP se.HHNEXP
## Non-cancer 127.35313 342.9523 12.347026 198.2589 571.98368 335.5276 134.30328
## Cancer    48.39815 254.3363  2.442972 106.6373  94.68786 1573.9158 39.09082
##          se.exp_total se.charges_total se.HHAMCR se.HHNMCR se.HHAMCD
## Non-cancer    353.2126      744.5026    153.401 13.708669 317.7343
## Cancer        1572.3906      1593.1331   1459.404  1.127506 634.7353
##          se.HHNMCD se.HHAPRV se.HHNPRV se.HHASLF se.HHNSLF
## Non-cancer 28.05631 84.25648 4.774802 54.87599 124.01835
## Cancer    25.74256 81.02720 1.685876 40.04733 25.75326
```

```
# National totals (pooled) among HHC users by cancer status
exp_totals_cn <- svyby(
  ~HHAEXP + HHNEXP + exp_total + charges_total +
    HHAMCR + HHNMCR + HHAMCD + HHNMCD + HHAPRV + HHNPRV + HHASLF + HHNSLF,
  ~factor(cancerpt, labels=c("Non-cancer", "Cancer")),
  design = users_design_all, svytotal, na.rm = TRUE)
exp_totals_cn
```

```
##          factor(cancerpt, labels = c("Non-cancer", "Cancer"))    HHAEXP
## Non-cancer                                                    Non-cancer 53665942653
## Cancer                                                         Cancer 16781528035
##          HHNEXP exp_total charges_total    HHAMCR    HHNMCR
## Non-cancer 5455027308 59120969961 72258092637 20347988566 134902264
## Cancer    304368446 17085896481 19779667924 9359121002 3018418
##          HHAMCD HHNMCD HHAPRV HHNPRV HHASLF HHNSLF
## Non-cancer 24723417544 902907520 2431461694 87537883 1405614837 4055246868
## Cancer    5414476297 88318766 464122428 4458027 194595776 172789961
##          se.HHAEXP se.HHNEXP se.exp_total se.charges_total se.HHAMCR
## Non-cancer 2752854826 964752172 2940527429 5664962352 1233279500
## Cancer    3123224577 72051325 3125959815 3213187743 2771385513
##          se.HHNMCR se.HHAMCD se.HHNMCD se.HHAPRV se.HHNPRV se.HHASLF
## Non-cancer 97355275 2326946034 199112778 600936230 33955402 389104678
## Cancer    2060835 1200377913 46966725 151200922 3079716 72725773
##          se.HHNSLF
## Non-cancer 888603910
## Cancer    47345753
```

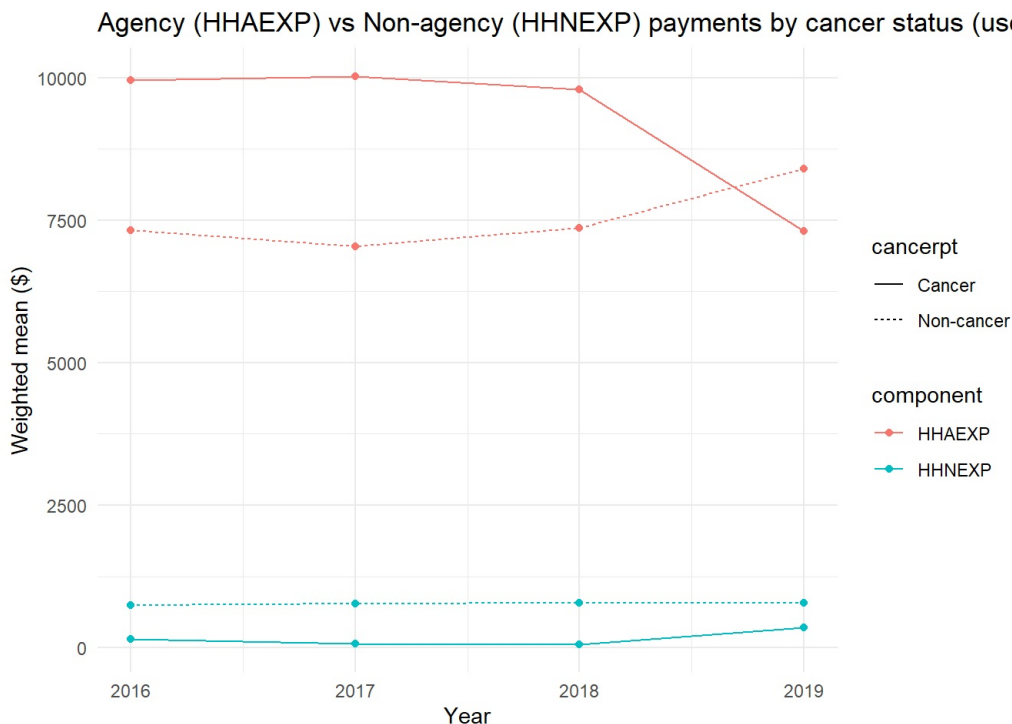
```
# YEARLY means among HHC users by cancer status
exp_year_cn_components <- svyby(
  ~HHAEXP + HHNEXP + charges_total + exp_total,
  ~interaction(year, cancerpt, drop=TRUE),
  design = users_design_all, svymean, na.rm = TRUE)
exp_year_cn_components
```



```
##      interaction(year, cancerpt, drop = TRUE)      HHAEXP      HHNEXP
## 2016.0      2016.0 7316.765 743.35094
## 2017.0      2017.0 7032.721 769.18139
## 2018.0      2018.0 7361.982 779.33965
## 2019.0      2019.0 8401.889 779.51801
## 2016.1      2016.1 9947.612 149.15560
## 2017.1      2017.1 10026.033 69.28508
## 2018.1      2018.1 9793.540 62.75560
## 2019.1      2019.1 7302.270 356.49001
##      charges_total exp_total se.HHAEXP se.HHNEXP se.charges_total
## 2016.0      9499.965 8060.116 499.7065 243.40105      643.4001
## 2017.0      9420.509 7801.903 542.6966 231.44659      679.0722
## 2018.0      9344.324 8141.321 669.1642 295.00316      767.9972
## 2019.0     12148.268 9181.407 804.3579 275.80592     2394.0431
## 2016.1     11346.792 10096.768 5410.0879 54.69258     5418.2454
## 2017.1     12373.005 10095.318 2197.3991 33.56399     2396.1301
## 2018.1     11315.281 9856.296 2031.7597 23.79262     2071.5790
## 2019.1      8768.141 7658.760 848.9287 127.80675      910.6164
##      se.exp_total
## 2016.0      548.0409
## 2017.0      553.3954
## 2018.0      711.5916
## 2019.0      839.9064
## 2016.1     5405.0156
## 2017.1     2193.5233
## 2018.1     2029.8423
## 2019.1      858.2836
```

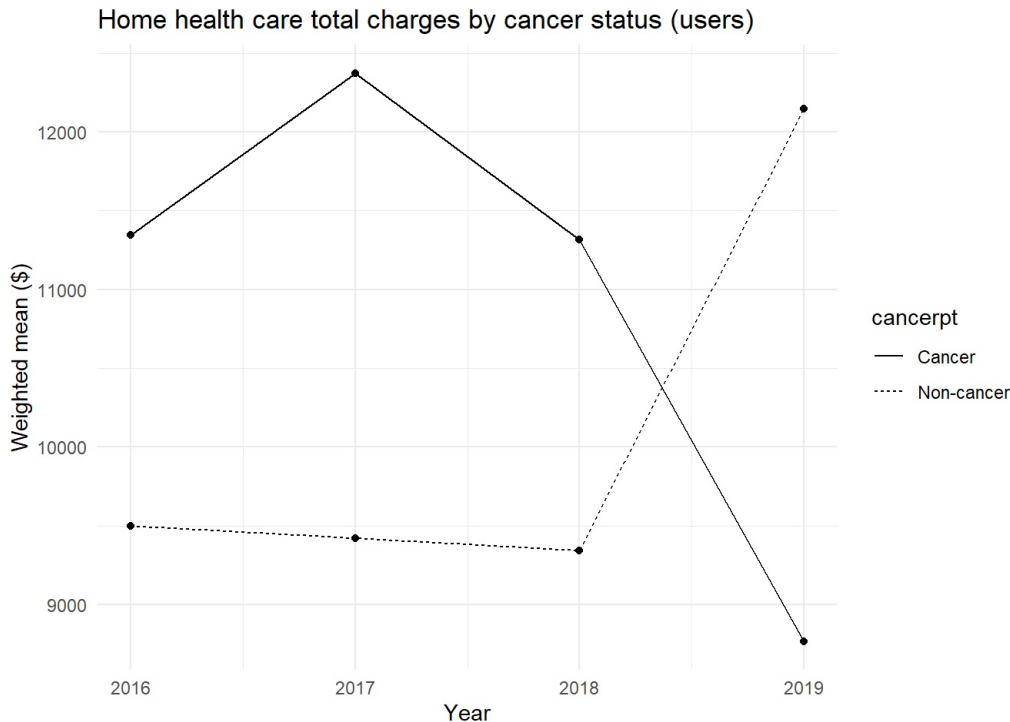
```
# Trend plot: Agency vs Non-agency payments by cancer status
suppressPackageStartupMessages({library(tidyr); library(dplyr); library(ggplot2)})
exp_long_comp <- exp_year_cn_components |>
  dplyr::rename(group = `interaction(year, cancerpt, drop = TRUE)`) |>
  tidyr::separate(group, into = c("year", "cancerpt"), sep="\\.\\.", convert = TRUE) |>
  dplyr::mutate(cancerpt = ifelse(cancerpt=="0", "Non-cancer", "Cancer")) |>
  dplyr::select(year, cancerpt, HHAEXP, HHNEXP) |>
  tidyr::pivot_longer(cols = c(HHAEXP, HHNEXP), names_to = "component", values_to = "mean")

ggplot(exp_long_comp, aes(x = year, y = mean, color = component, linetype = cancerpt)) +
  geom_line() + geom_point() +
  labs(x = "Year", y = "Weighted mean ($)"),
  title = "Agency (HHAEXP) vs Non-agency (HHNEXP) payments by cancer status (users)" +
  theme_minimal()
```



```
# Trend plot: Total charges (HHATCH + HHNTCH)
exp_long_chg <- exp_year_cn_components |>
  dplyr::rename(group = `interaction(year, cancerpt, drop = TRUE)`) |>
  tidyr::separate(group, into = c("year","cancerpt"), sep="\\.", convert = TRUE) |>
  dplyr::mutate(cancerpt = ifelse(cancerpt=="0","Non-cancer","Cancer")) |>
  dplyr::select(year, cancerpt, charges_total)

ggplot(exp_long_chg, aes(x = year, y = charges_total, linetype = cancerpt)) +
  geom_line() + geom_point() +
  labs(x = "Year", y = "Weighted mean ($)",
       title = "Home health care total charges by cancer status (users)") +
  theme_minimal()
```

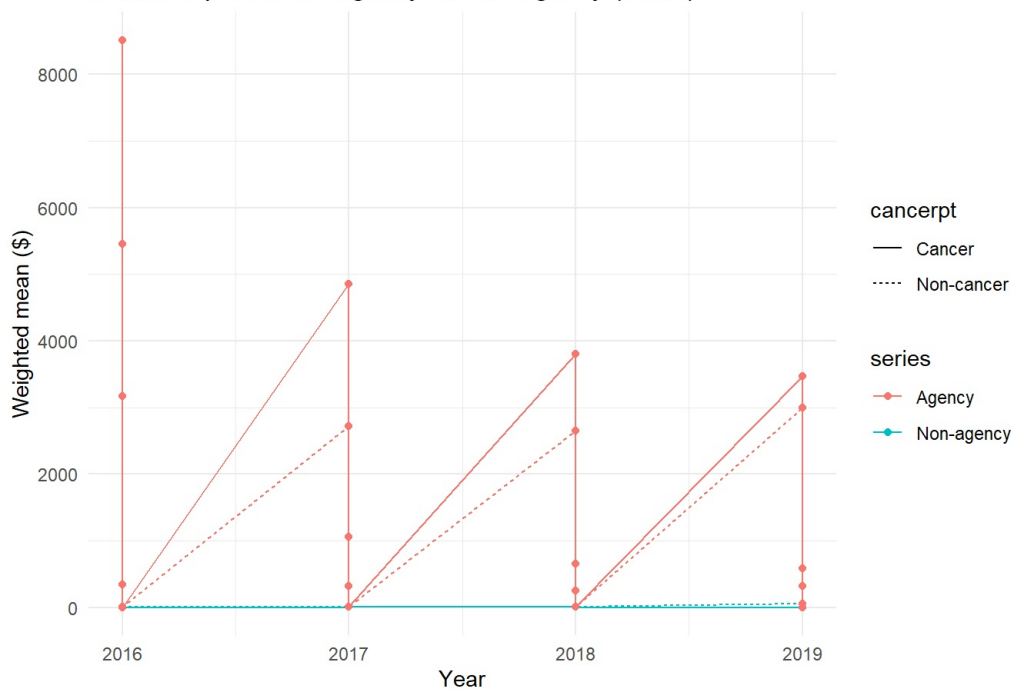


```
# Payer split: agency vs non-agency components by year x cancer
payer_year_cn <- svyby(
  ~HHAMCR + HHNMCR + HHAMCD + HHNMCD + HHAPRV + HHNPRV + HHASLF + HHNSLF,
  ~interaction(year, cancerpt, drop=TRUE),
  design = users_design_all, svymean, na.rm = TRUE)

payer_long <- payer_year_cn |>
  dplyr::rename(group = `interaction(year, cancerpt, drop = TRUE)`) |>
  tidyr::separate(group, into = c("year","cancerpt"), sep="\\.", convert = TRUE) |>
  dplyr::mutate(cancerpt = ifelse(cancerpt=="0","Non-cancer","Cancer")) |>
  tidyr::pivot_longer(-c(year,cancerpt), names_to = "series", values_to = "mean") |>
  dplyr::mutate(
    payer = dplyr::case_when(grepl("MCR$", series) ~ "Medicare",
                             grepl("MCD$", series) ~ "Medicaid",
                             grepl("PRV$", series) ~ "Private",
                             grepl("SLF$", series) ~ "OOP",
                             TRUE ~ "Other"),
    setting = ifelse(grepl("^HHN", series), "Non-agency", "Agency")
  )

# Example visualization: Medicare-paid means, agency vs non-agency
medicare_plot <- payer_long |>
  dplyr::filter(payer == "Medicare") |>
  dplyr::mutate(series = ifelse(grepl("^HHN", series), "Non-agency", "Agency")) |>
  ggplot(aes(x = year, y = mean, color = series, linetype = cancerpt)) +
  geom_line() + geom_point() +
  labs(x = "Year", y = "Weighted mean ($)", title = "Medicare-paid HHC: Agency vs Non-agency (users)") +
  theme_minimal()
medicare_plot
```

Medicare-paid HHC: Agency vs Non-agency (users)



Models

1. Utilization (any home care) — logistic, stratified by cancer status

```
library(knitr)
# library(kableExtra) # uncomment for styling

# helper so you don't repeat yourself
tidy_kable <- function(model, caption, exponentiate = FALSE, digits = 3) {
  broom::tidy(model, conf.int = TRUE, exponentiate = exponentiate) |>
    knitr::kable(digits = digits,
                  caption = caption,
                  format.args = list(big.mark = ","))
  # |> kableExtra::kable_styling(full_width = FALSE) # if using kableExtra
}
use_cancer <- svyglm(homecare_any ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEQ + MARRY,
                     design = subset(meps_design, cancerpt == 1), family = quasibinomial())
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
## Warning in summary.glm(glm.object): observations with zero weight not used for
## calculating dispersion
```

```
use_noncancer <- svyglm(homecare_any ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEQ + MARRY,
                       design = subset(meps_design, cancerpt == 0), family = quasibinomial())
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
tidy_kable(use_cancer, "Utilization (logit): Cancer — adjusted ORs", exponentiate = TRUE)
```

Utilization (logit): Cancer — adjusted ORs

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.004	0.427	-12.755	0.000	0.002	0.010
AGE	1.059	0.006	9.561	0.000	1.046	1.071
SEXFemale	1.158	0.101	1.441	0.150	0.948	1.413
RACEBlack	1.004	0.150	0.504	0.555	0.814	1.170

RACEBlack	1.094	0.152	0.591	0.555	0.811	1.476
RACENative	0.984	0.577	-0.027	0.978	0.317	3.057
RACEAsianPI	0.486	0.376	-1.924	0.055	0.232	1.015
RACEMultiple	1.987	0.292	2.353	0.019	1.120	3.523
INSCOVPublicOnly	1.102	0.106	0.909	0.364	0.894	1.357
INSCOVUninsured	0.365	0.689	-1.462	0.144	0.094	1.413
POVCATNearPoor	0.743	0.216	-1.381	0.168	0.486	1.134
POVCATLowIncome	0.767	0.167	-1.591	0.112	0.552	1.064
POVCATMiddle	0.554	0.158	-3.741	0.000	0.406	0.755
POVCATHigh	0.358	0.174	-5.904	0.000	0.254	0.503
HIDEGGED	0.762	0.240	-1.131	0.258	0.475	1.221
HIDEGHS	0.584	0.128	-4.195	0.000	0.454	0.751
HIDEGSomeCollege	0.653	0.185	-2.309	0.021	0.454	0.938
HIDEGAA	0.425	0.246	-3.476	0.001	0.262	0.689
HIDEGBA	0.431	0.350	-2.406	0.016	0.216	0.856
HIDEGMA	0.551	0.209	-2.848	0.005	0.366	0.831
MARRYWidowed	1.467	0.146	2.631	0.009	1.102	1.953
MARRYDivorced	1.323	0.154	1.816	0.070	0.978	1.791
MARRYSeparated	1.093	0.354	0.250	0.803	0.545	2.191
MARRYNeverMarried	2.344	0.186	4.593	0.000	1.629	3.374

tidy_kable(use_noncancer, "Utilization (logit): Non-cancer – adjusted ORs", exponentiate = TRUE)

Utilization (logit): Non-cancer — adjusted ORs

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.001	0.158	-44.355	0.000	0.001	0.001
AGE	1.064	0.002	30.538	0.000	1.060	1.068
SEXFemale	1.253	0.051	4.418	0.000	1.133	1.385
RACEBlack	1.078	0.064	1.169	0.243	0.950	1.224
RACENative	1.066	0.224	0.285	0.776	0.687	1.655
RACEAsianPI	0.775	0.138	-1.841	0.066	0.591	1.017
RACEMultiple	1.324	0.137	2.049	0.041	1.012	1.733
INSCOVPublicOnly	2.051	0.054	13.211	0.000	1.843	2.282
INSCOVUninsured	0.308	0.227	-5.181	0.000	0.197	0.481
POVCATNearPoor	0.773	0.093	-2.773	0.006	0.644	0.927
POVCATLowIncome	0.678	0.072	-5.371	0.000	0.589	0.782
POVCATMiddle	0.533	0.075	-8.376	0.000	0.459	0.617
POVCATHigh	0.463	0.083	-9.231	0.000	0.394	0.546
HIDEGGED	1.184	0.108	1.563	0.118	0.958	1.463
HIDEGHS	0.852	0.065	-2.460	0.014	0.750	0.968
HIDEGSomeCollege	0.719	0.097	-3.386	0.001	0.594	0.870
HIDEGAA	0.763	0.121	-2.241	0.025	0.601	0.967
HIDEGBA	0.819	0.182	-1.102	0.271	0.573	1.169
HIDEGMA	0.851	0.103	-1.578	0.115	0.696	1.040
MARRYWidowed	2.054	0.076	9.533	0.000	1.771	2.383
MARRYDivorced	1.532	0.070	6.075	0.000	1.335	1.759
MARRYSeparated	2.301	0.124	6.728	0.000	1.805	2.935

MARRYNeverMarried

2.606

0.083

11.591

0.000

2.216

3.065

Optional formal test of differences: fit an interaction model

```
use_interact <- svyglm(homecare_any ~ cancerpt*(AGE + SEX + RACE + INSCOV + POVCAT + HIDEQ + MARRY),  
  design = meps_design, family = quasibinomial())
```

```
## Warning in summary.glm(g): observations with zero weight not used for  
## calculating dispersion
```

```
## Warning in summary.glm(glm.object): observations with zero weight not used for  
## calculating dispersion
```

```
summary(use_interact)
```

```
##
## Call:
## svyglm(formula = homedata ~ cancerpt * (AGE + SEX + RACE +
##       INSCOV + POVCAT + HIDEG + MARRY), design = meps_design, family = quasibinomial())
##
## Survey design:
## update(meps_design, charges_total = ifelse(is.na(HHATCH), 0,
##       HHATCH) + ifelse(is.na(HHNTCH), 0, HHNTCH))
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.011976    0.158086  -44.355 < 2e-16 ***
## cancerpt        1.564687    0.450997   3.469 0.000549 ***
## AGE            0.061835    0.002025  30.538 < 2e-16 ***
## SEXFemale      0.225424    0.051027   4.418 1.13e-05 ***
## RACEBlack      0.075296    0.064420   1.169 0.242815
## RACENative     0.063962    0.224224   0.285 0.775519
## RACEAsianPI    -0.254606    0.138320  -1.841 0.066029 .
## RACEMultiple   0.280796    0.137063   2.049 0.040816 *
## INSCOVPublicOnly 0.718203    0.054365  13.211 < 2e-16 ***
## INSCOVUninsured -1.176865    0.227164  -5.181 2.79e-07 ***
## POVCATNearPoor -0.258011    0.093039  -2.773 0.005679 **
## POVCATLowIncome -0.388245    0.072285  -5.371 1.02e-07 ***
## POVCATMiddle   -0.630161    0.075235  -8.376 2.39e-16 ***
## POVCATHigh     -0.768959    0.083299  -9.231 < 2e-16 ***
## HIDEGGED       0.168750    0.107955   1.563 0.118405
## HIDEGHS        -0.159813    0.064964  -2.460 0.014099 *
## HIDEGSomeCollege -0.329948    0.097435  -3.386 0.000742 ***
## HIDEGAA        -0.271035    0.120965  -2.241 0.025320 *
## HIDEGBA        -0.200085    0.181567  -1.102 0.270791
## HIDEGMA        -0.161849    0.102538  -1.578 0.114855
## MARRYWidowed   0.719952    0.075519   9.533 < 2e-16 ***
## MARRYDivorced  0.426715    0.070236   6.075 1.90e-09 ***
## MARRYSeparated 0.833537    0.123895   6.728 3.25e-11 ***
## MARRYNeverMarried 0.957765    0.082628  11.591 < 2e-16 ***
## cancerpt:AGE   -0.004760    0.006212  -0.766 0.443720
## cancerpt:SEXFemale -0.079159    0.117080  -0.676 0.499161
## cancerpt:RACEBlack 0.014733    0.160291   0.092 0.926789
## cancerpt:RACENative -0.079744    0.601741  -0.133 0.894604
## cancerpt:RACEAsianPI -0.467923    0.395196  -1.184 0.236749
## cancerpt:RACEMultiple 0.405664    0.305775   1.327 0.184989
## cancerpt:INSCOVPublicOnly -0.621484    0.116811  -5.320 1.34e-07 ***
## cancerpt:INSCOVUninsured 0.169437    0.724250   0.234 0.815084
## cancerpt:POVCATNearPoor -0.039666    0.236116  -0.168 0.866630
## cancerpt:POVCATLowIncome 0.122377    0.180001   0.680 0.496783
## cancerpt:POVCATMiddle 0.039285    0.172686   0.227 0.820097
## cancerpt:POVCATHigh -0.259175    0.192773  -1.344 0.179175
## cancerpt:HIDEGGED -0.440853    0.270971  -1.627 0.104138
## cancerpt:HIDEGHS -0.377725    0.145865  -2.590 0.009782 **
## cancerpt:HIDEGSomeCollege -0.096990    0.213679  -0.454 0.650019
## cancerpt:HIDEGAA -0.584687    0.280786  -2.082 0.037625 *
## cancerpt:HIDEGBA -0.642578    0.390025  -1.648 0.099835 .
## cancerpt:HIDEGMA -0.433362    0.234492  -1.848 0.064953 .
## cancerpt:MARRYWidowed -0.336632    0.164937  -2.041 0.041576 *
## cancerpt:MARRYDivorced -0.146738    0.165018  -0.889 0.374146
## cancerpt:MARRYSeparated -0.745014    0.378926  -1.966 0.049625 *
## cancerpt:MARRYNeverMarried -0.105745    0.201268  -0.525 0.599453
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9912099)
##
## Number of Fisher Scoring iterations: 8
```

2. Intensity (days among users) — quasi-Poisson, stratified by cancer status

```
days_cancer <- svyglm(HHTOTD ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEG + MARRY,
  design = subset(meps_design, cancerpt == 1 & homedata == 1),
  family = quasipoisson(link = "log"))
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
## Warning in summary.glm(glm.object): observations with zero weight not used for
## calculating dispersion
```

```
days_noncancer <- svyglm(HHTOTD ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEG + MARRY,
  design = subset(meps_design, cancerpt == 0 & homecare_any == 1),
  family = quasipoisson(link = "log"))
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
tidy_kable(days_cancer, "Intensity (quasi-Poisson): Cancer – adjusted rate ratios", exponentiate = TRUE)
```

Intensity (quasi-Poisson): Cancer — adjusted rate ratios

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	34.140	0.459	7.699	0.000	13.763	84.690
AGE	1.016	0.006	2.831	0.005	1.005	1.028
SEXFemale	1.032	0.139	0.227	0.821	0.783	1.360
RACEBlack	1.117	0.147	0.752	0.453	0.834	1.496
RACENative	1.297	0.435	0.598	0.551	0.548	3.070
RACEAsianPI	0.211	0.469	-3.313	0.001	0.083	0.535
RACEMultiple	1.641	0.279	1.773	0.079	0.943	2.854
INSCOVPublicOnly	1.687	0.195	2.686	0.008	1.147	2.482
INSCOVUninsured	0.345	0.265	-4.024	0.000	0.204	0.582
POVCATNearPoor	1.200	0.219	0.833	0.407	0.778	1.850
POVCATLowIncome	0.785	0.220	-1.102	0.273	0.507	1.214
POVCATMiddle	0.606	0.211	-2.371	0.019	0.399	0.921
POVCATHigh	0.638	0.207	-2.173	0.032	0.423	0.961
HIDEGGED	0.588	0.246	-2.158	0.033	0.362	0.957
HIDEGHS	0.613	0.172	-2.840	0.005	0.435	0.862
HIDEGSomeCollege	0.729	0.231	-1.366	0.175	0.461	1.153
HIDEGAA	0.631	0.363	-1.267	0.208	0.307	1.296
HIDEGBA	0.207	0.401	-3.933	0.000	0.094	0.458
HIDEGMA	0.744	0.416	-0.711	0.479	0.326	1.697
MARRYWidowed	0.825	0.225	-0.858	0.393	0.528	1.287
MARRYDivorced	0.655	0.233	-1.819	0.072	0.413	1.039
MARRYSeparated	0.506	0.560	-1.214	0.227	0.167	1.537
MARRYNeverMarried	1.230	0.270	0.768	0.444	0.721	2.101

```
tidy_kable(days_noncancer, "Intensity (quasi-Poisson): Non-cancer – adjusted rate ratios", exponentiate = TRUE)
```

Intensity (quasi-Poisson): Non-cancer — adjusted rate ratios

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	57.600	0.192	21.139	0.000	39.517	83.958
AGE	1.002	0.002	1.105	0.270	0.998	1.007
SEXFemale	0.899	0.060	-1.768	0.078	0.800	1.012
RACEBlack	1.232	0.062	3.338	0.001	1.089	1.392
RACENative	1.338	0.232	1.256	0.210	0.848	2.111
RACEAsianPI	1.040	0.151	0.262	0.794	0.774	1.398

RACEMultiple	0.990	0.136	-0.071	0.944	0.758	1.294
INSCOVPublicOnly	1.443	0.084	4.387	0.000	1.224	1.700
INSCOVUninsured	0.493	0.306	-2.316	0.021	0.270	0.898
POVCATNearPoor	0.881	0.092	-1.368	0.172	0.735	1.057
POVCATLowIncome	0.926	0.076	-1.011	0.312	0.797	1.075
POVCATMiddle	0.814	0.086	-2.377	0.018	0.687	0.965
POVCATHigh	0.763	0.099	-2.725	0.007	0.628	0.927
HIDEGGED	0.573	0.117	-4.746	0.000	0.454	0.721
HIDEGHS	0.733	0.069	-4.512	0.000	0.640	0.839
HIDEGSomeCollege	0.793	0.126	-1.844	0.066	0.619	1.015
HIDEGAA	0.556	0.175	-3.354	0.001	0.394	0.784
HIDEGBA	0.538	0.265	-2.343	0.020	0.319	0.905
HIDEGMA	0.689	0.120	-3.109	0.002	0.544	0.872
MARRYWidowed	1.666	0.091	5.631	0.000	1.394	1.990
MARRYDivorced	1.486	0.089	4.455	0.000	1.248	1.769
MARRYSeparated	1.524	0.131	3.204	0.001	1.177	1.972
MARRYNeverMarried	1.882	0.098	6.460	0.000	1.553	2.282

3. Costs (log total expenditures among users) — linear, **stratified by cancer status**

```
cost_cancer <- svyglm(log(exp_total + 1) ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEG + MARRY,
  design = subset(meps_design, cancerpt == 1 & exp_total > 0))
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
## Warning in summary.glm(glm.object): observations with zero weight not used for
## calculating dispersion
```

```
cost_noncancer <- svyglm(log(exp_total + 1) ~ AGE + SEX + RACE + INSCOV + POVCAT + HIDEG + MARRY,
  design = subset(meps_design, cancerpt == 0 & exp_total > 0))
```

```
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
## Warning in summary.glm(g): observations with zero weight not used for
## calculating dispersion
```

```
# COSTS (log-linear): coefficients on log(exp_total + 1)
tidy_kable(cost_cancer, "Costs (log-linear): Cancer — full results", exponentiate = FALSE)
```

Costs (log-linear): Cancer — full results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	6.435	0.492	13.075	0.000	5.459	7.410
AGE	0.025	0.006	4.107	0.000	0.013	0.037
SEXFemale	0.147	0.145	1.016	0.312	-0.140	0.434
RACEBlack	0.157	0.198	0.793	0.430	-0.236	0.550
RACENative	-0.263	0.714	-0.368	0.714	-1.679	1.153
RACEAsianPI	0.045	0.257	0.176	0.861	-0.465	0.556
RACEMultiple	0.137	0.394	0.348	0.729	-0.644	0.919
INSCOVPublicOnly	0.608	0.181	3.354	0.001	0.249	0.967
INSCOVUninsured	-1.588	0.262	-6.072	0.000	-2.106	-1.070
POVCATNearPoor	0.403	0.314	1.284	0.202	-0.219	1.025

POVCATLowIncome	-0.315	0.276	-1.142	0.256	-0.861	0.232
POVCATMiddle	-0.339	0.196	-1.730	0.087	-0.728	0.050
POVCATHigh	-0.671	0.240	-2.793	0.006	-1.148	-0.195
HIDEGGED	-0.030	0.261	-0.114	0.909	-0.547	0.488
HIDEGHS	-0.371	0.183	-2.028	0.045	-0.734	-0.008
HIDEGSomeCollege	-0.333	0.241	-1.380	0.170	-0.811	0.145
HIDEGAA	-0.336	0.455	-0.737	0.462	-1.237	0.566
HIDEGBA	0.166	0.432	0.385	0.701	-0.690	1.022
HIDEGMA	-0.144	0.388	-0.371	0.712	-0.913	0.625
MARRYWidowed	-0.144	0.205	-0.701	0.485	-0.551	0.263
MARRYDivorced	-0.442	0.237	-1.864	0.065	-0.913	0.028
MARRYSeparated	-0.658	0.572	-1.151	0.252	-1.792	0.476
MARRYNeverMarried	0.138	0.331	0.418	0.677	-0.518	0.794

```
tidy_kable(cost_noncancer, "Costs (log-linear): Non-cancer – full results", exponentiate = FALSE)
```

Costs (log-linear): Non-cancer — full results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	7.119	0.239	29.839	0.000	6.650	7.587
AGE	0.019	0.003	6.894	0.000	0.013	0.024
SEXFemale	-0.200	0.076	-2.646	0.008	-0.349	-0.052
RACEBlack	0.124	0.088	1.405	0.161	-0.049	0.296
RACENative	0.476	0.257	1.852	0.065	-0.029	0.981
RACEAsianPI	0.185	0.212	0.873	0.383	-0.231	0.600
RACEMultiple	0.253	0.162	1.559	0.120	-0.066	0.571
INSCOVPublicOnly	0.360	0.085	4.219	0.000	0.192	0.527
INSCOVUninsured	-0.428	0.394	-1.087	0.278	-1.202	0.346
POVCATNearPoor	-0.328	0.122	-2.691	0.007	-0.567	-0.088
POVCATLowIncome	-0.316	0.110	-2.885	0.004	-0.531	-0.101
POVCATMiddle	-0.527	0.101	-5.214	0.000	-0.725	-0.328
POVCATHigh	-0.521	0.119	-4.376	0.000	-0.754	-0.287
HIDEGGED	-0.495	0.155	-3.185	0.002	-0.800	-0.190
HIDEGHS	-0.292	0.092	-3.155	0.002	-0.473	-0.110
HIDEGSomeCollege	-0.306	0.137	-2.224	0.027	-0.576	-0.036
HIDEGAA	-0.533	0.180	-2.964	0.003	-0.886	-0.180
HIDEGBA	-0.514	0.356	-1.447	0.149	-1.213	0.184
HIDEGMA	-0.432	0.150	-2.877	0.004	-0.726	-0.137
MARRYWidowed	0.241	0.093	2.589	0.010	0.058	0.423
MARRYDivorced	0.239	0.106	2.249	0.025	0.030	0.448
MARRYSeparated	0.183	0.194	0.944	0.346	-0.198	0.565
MARRYNeverMarried	0.618	0.121	5.093	0.000	0.379	0.856

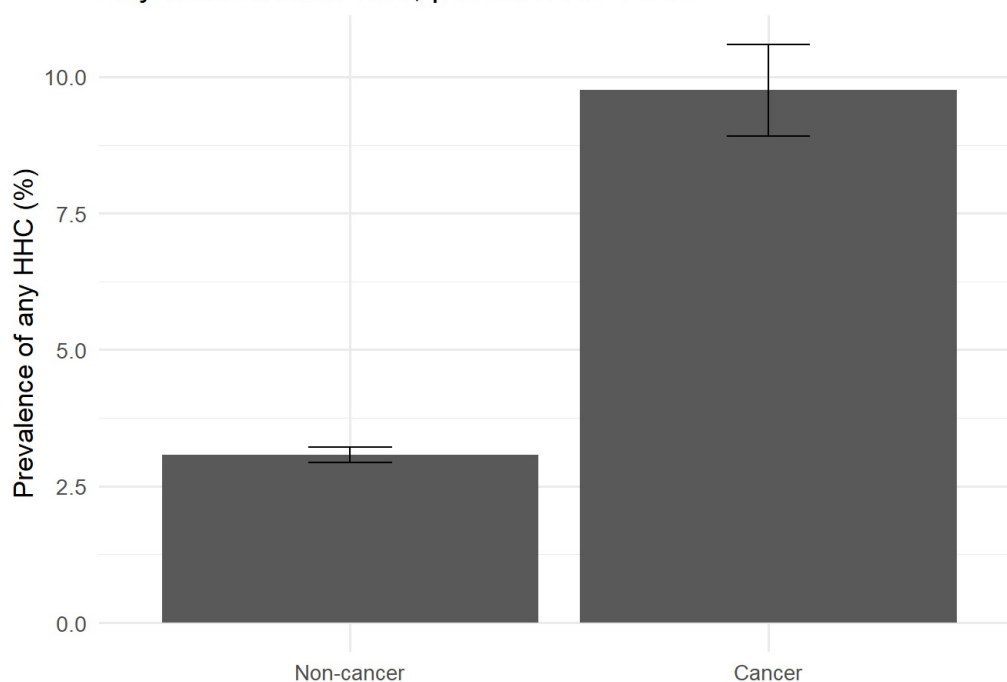
Additional visualizations

Prevalence — overview & by subgroups

```
# Overall prevalence (% any HHC) by cancer status
prev_overall <- svyby(~homecare_any,
  ~factor(cancerpt, labels=c("Non-cancer", "Cancer")),
  design = meps_design, svymean)
prev_overall <- prev_overall %>%
  as_tibble() %>%
  dplyr::rename(cancer = `factor(cancerpt, labels = c("Non-cancer", "Cancer"))`) %>%
  dplyr::mutate(pct = 100*homecare_any,
    se = 100*se,
    lwr = pct - 1.96*se,
    upr = pct + 1.96*se)

ggplot(prev_overall, aes(x = cancer, y = pct)) +
  geom_col() +
  geom_errorbar(aes(ymin = lwr, ymax = upr), width = 0.2) +
  labs(x = NULL, y = "Prevalence of any HHC (%)",
    title = "Any home health care, pooled 2016–2019") +
  theme_minimal(base_size = 13)
```

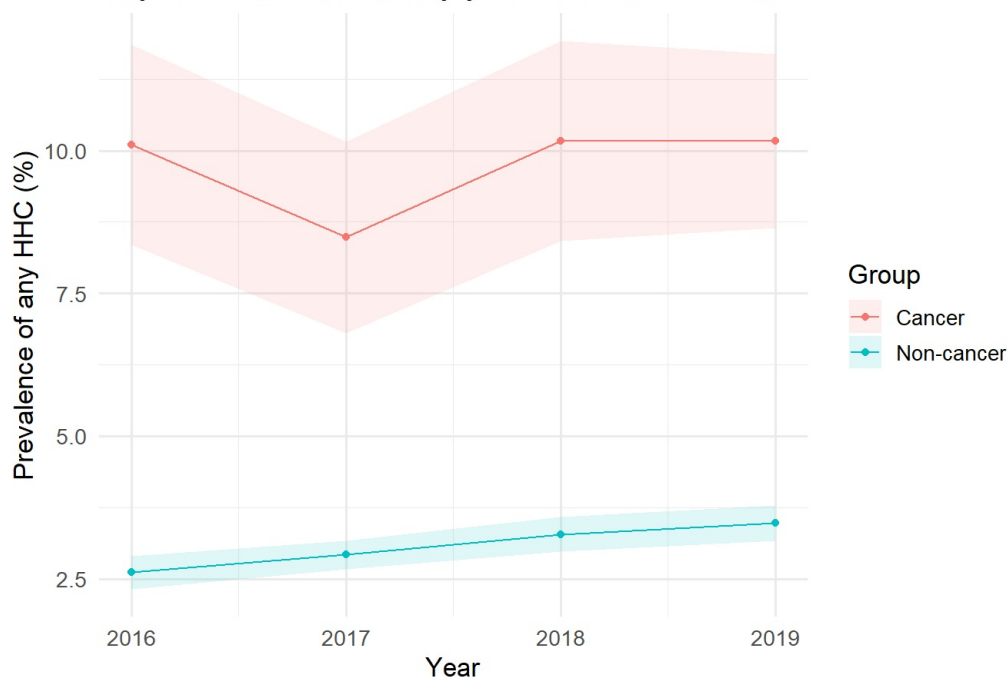
Any home health care, pooled 2016–2019



```
# Yearly prevalence by cancer status
prev_year_cn <- svyby(~homecare_any,
  ~interaction(year, cancerpt, drop=TRUE),
  design = meps_design, svymean, na.rm = TRUE)
prev_year_cn <- prev_year_cn %>%
  rename(group = `interaction(year, cancerpt, drop = TRUE)`) %>%
  tidyr::separate(group, into=c("year", "cancerpt"), sep="\\. ", convert=TRUE) %>%
  mutate(cancerpt = ifelse(cancerpt=="0", "Non-cancer", "Cancer"),
    pct = 100*homecare_any,
    se = 100*se,
    lwr = pct - 1.96*se,
    upr = pct + 1.96*se)

ggplot(prev_year_cn, aes(x = year, y = pct, color = cancerpt))+
  geom_line() + geom_point() +
  geom_ribbon(aes(ymin=lwr, ymax=upr, fill=cancerpt), alpha=0.12, color=NA) +
  labs(x = "Year", y = "Prevalence of any HHC (%)",
    title = "Any home health care by year and cancer status",
    color = "Group", fill = "Group") +
  theme_minimal(base_size = 13)
```

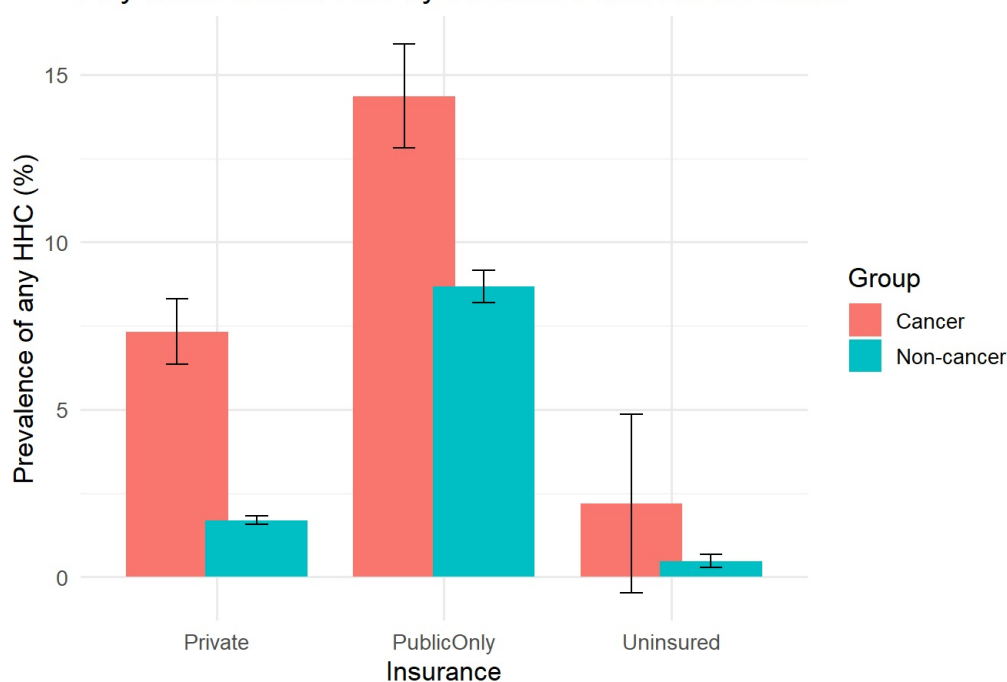
Any home health care by year and cancer status



```
# Prevalence by insurance, Cancer vs Non-cancer
prev_ins_c <- svyby(~homecare_any, ~INSCOV, design = subset(meps_design, cancerpt==1), svymean)
prev_ins_nc <- svyby(~homecare_any, ~INSCOV, design = subset(meps_design, cancerpt==0), svymean)
prev_ins <- bind_rows(
  prev_ins_c %>% mutate(cancer = "Cancer"),
  prev_ins_nc %>% mutate(cancer = "Non-cancer")
) %>%
  as_tibble() %>%
  mutate(pct = 100*homecare_any, se = 100*se,
         lwr = pct - 1.96*se, upr = pct + 1.96*se)

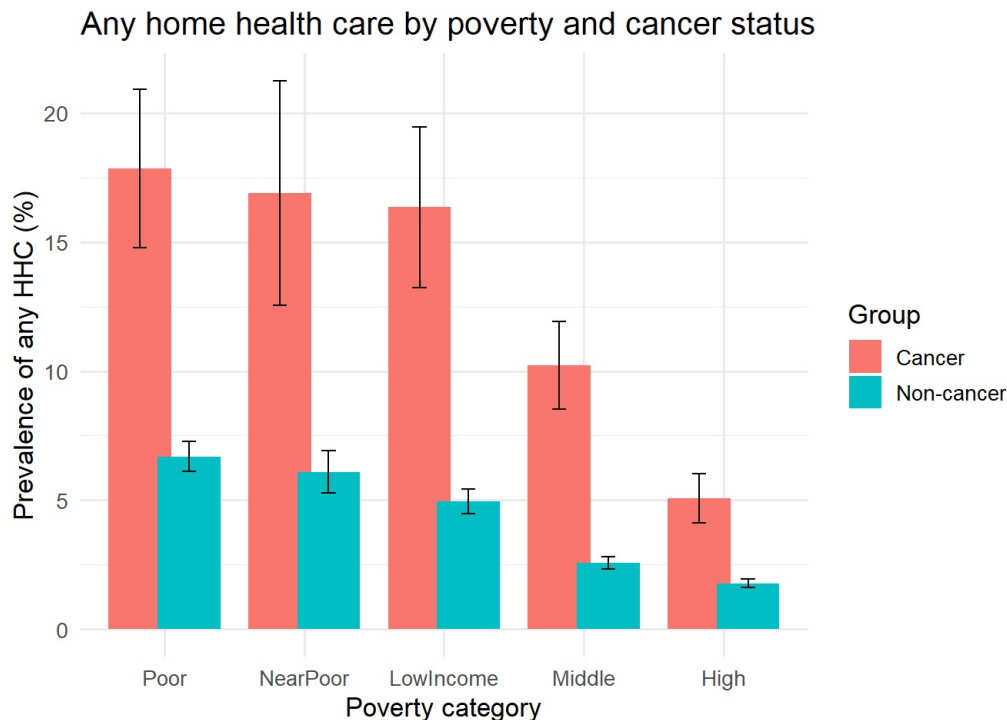
ggplot(prev_ins, aes(x = INSCOV, y = pct, fill = cancer)) +
  geom_col(position = position_dodge(width = 0.7)) +
  geom_errorbar(aes(ymin = lwr, ymax = upr), position = position_dodge(width = 0.7), width = 0.2) +
  labs(x = "Insurance", y = "Prevalence of any HHC (%)",
       title = "Any home health care by insurance and cancer status",
       fill = "Group") +
  theme_minimal(base_size = 13)
```

Any home health care by insurance and cancer status



```
# Prevalence by poverty category, Cancer vs Non-cancer
prev_pov_c <- svyby(~homocare_any, ~POVCAT, design = subset(meps_design, cancerpt==1), svymean)
prev_pov_nc <- svyby(~homocare_any, ~POVCAT, design = subset(meps_design, cancerpt==0), svymean)
prev_pov <- bind_rows(
  prev_pov_c %>% mutate(cancer = "Cancer"),
  prev_pov_nc %>% mutate(cancer = "Non-cancer")
) %>% as_tibble() %>%
  mutate(pct = 100*homocare_any, se = 100*se,
         lwr = pct - 1.96*se, upr = pct + 1.96*se)

ggplot(prev_pov, aes(x = POVCAT, y = pct, fill = cancer))+
  geom_col(position = position_dodge(width = 0.7))+
  geom_errorbar(aes(ymin = lwr, ymax = upr), position = position_dodge(width = 0.7), width = 0.2)+
  labs(x = "Poverty category", y = "Prevalence of any HHC (%)",
       title = "Any home health care by poverty and cancer status",
       fill = "Group") +
  theme_minimal(base_size = 13)
```

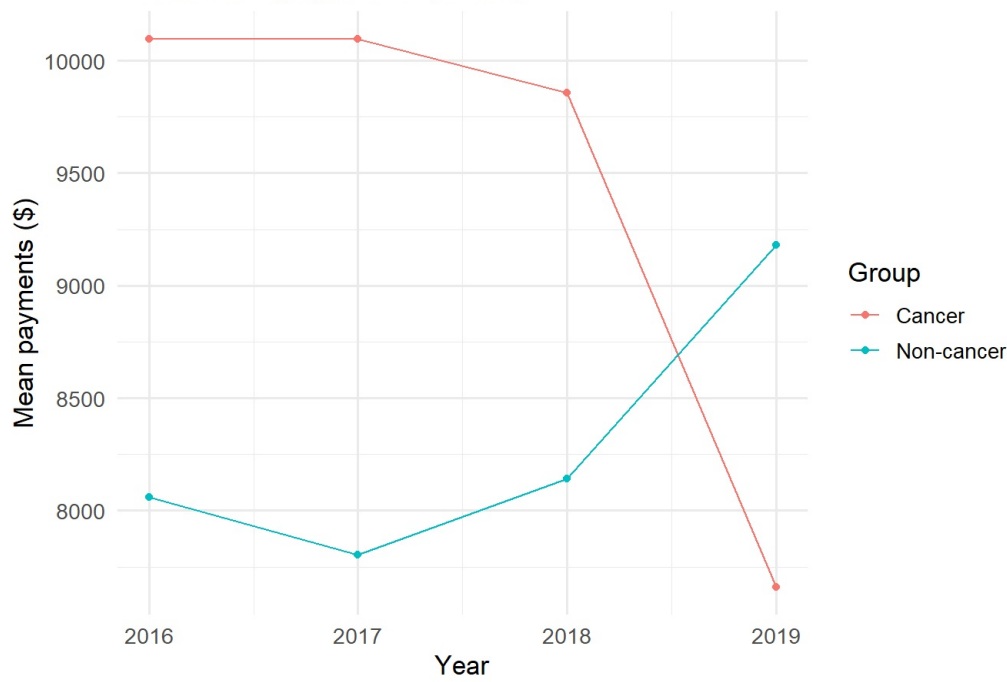


Expenditures — totals, payer mix, and charges vs. payments

```
# Total mean payments (HHAEXP+HHNEXP) by year & cancer status
exp_year_cn_tot <- svyby(~exp_total, ~interaction(year, cancerpt, drop=TRUE),
  design = users_design_all, svymean, na.rm = TRUE) %>%
  rename(group = `interaction(year, cancerpt, drop = TRUE)`) %>%
  tidyr::separate(group, into=c("year","cancerpt"), sep="\\. ", convert=TRUE) %>%
  mutate(cancerpt = ifelse(cancerpt=="0","Non-cancer","Cancer"))

exp_year_cn_tot %>%
  ggplot(aes(x = year, y = exp_total, color = cancerpt))+
  geom_line() + geom_point() +
  labs(x = "Year", y = "Mean payments ($)", color = "Group",
       title = "Total HHC payments (users)")+
  theme_minimal(base_size = 13)
```

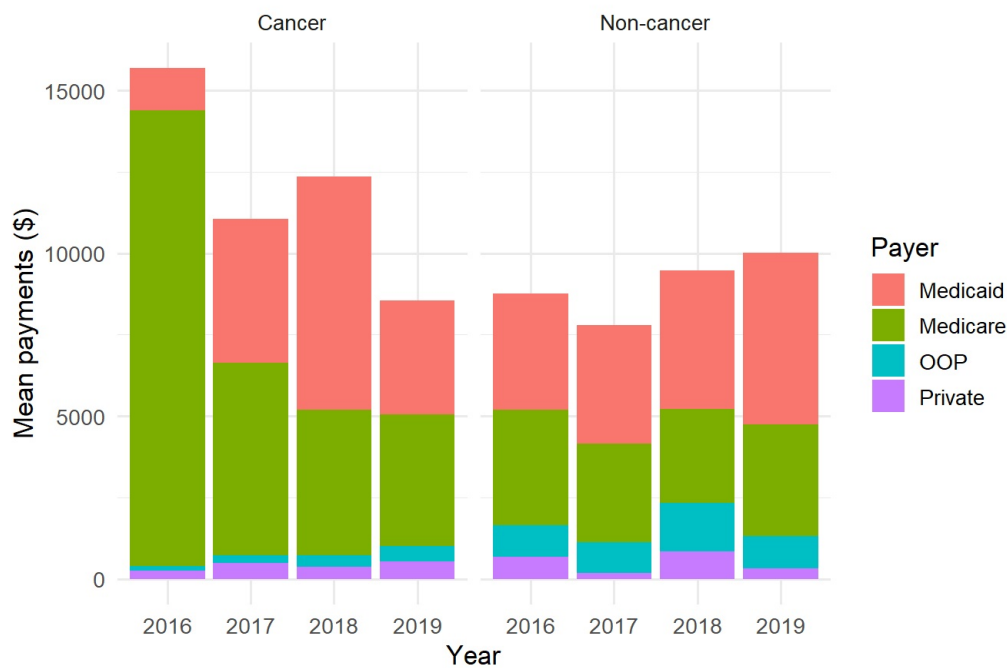
Total HHC payments (users)



```
# Payer composition stacked bars (summing agency + non-agency), by year & cancer status
payer_sum <- payer_long %>%
  group_by(year, cancerpt, payer) %>%
  summarise(mean = sum(mean), .groups = "drop")

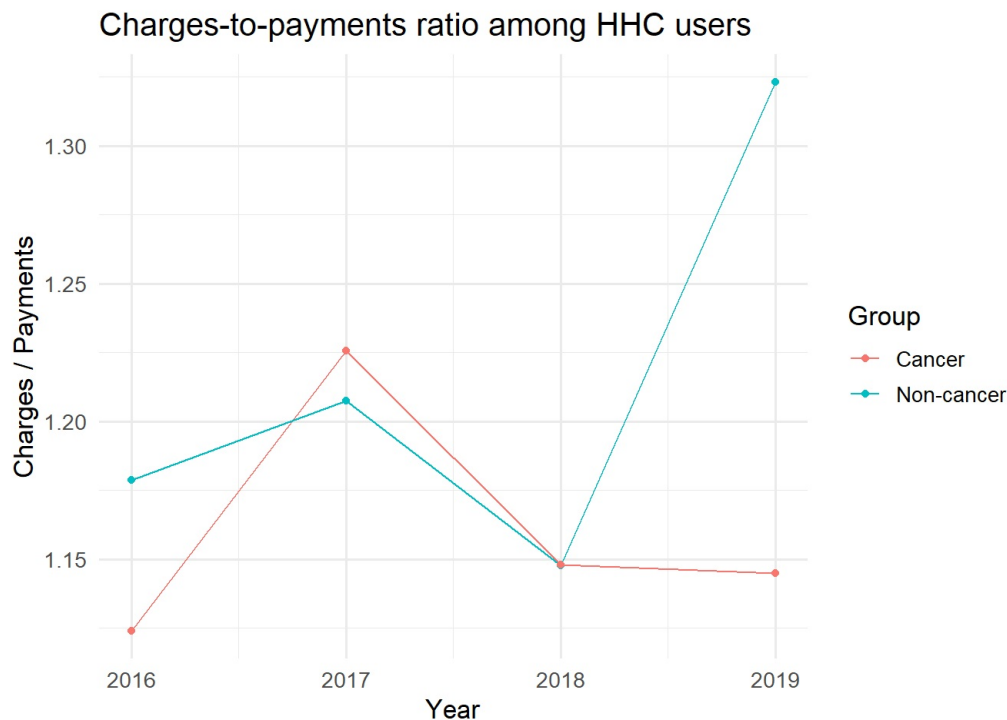
ggplot(payer_sum, aes(x = factor(year), y = mean, fill = payer)) +
  geom_col() +
  facet_wrap(~cancerpt) +
  labs(x = "Year", y = "Mean payments ($)", fill = "Payer",
       title = "Payer mix among HHC users, by cancer status") +
  theme_minimal(base_size = 13)
```

Payer mix among HHC users, by cancer status



```
# Charges-to-payments ratio by year & cancer status
ratio_df <- exp_year_cn_components %>%
  rename(group = `interaction(year, cancerpt, drop = TRUE)` ) %>%
  tidyr::separate(group, into=c("year","cancerpt"), sep="\\. ", convert=TRUE) %>%
  mutate(cancerpt = ifelse(cancerpt=="0","Non-cancer","Cancer"),
         ratio = charges_total/exp_total)

ggplot(ratio_df, aes(x = year, y = ratio, color = cancerpt))+
  geom_line() + geom_point() +
  labs(x = "Year", y = "Charges / Payments",
       color = "Group",
       title = "Charges-to-payments ratio among HHC users")+
  theme_minimal(base_size = 13)
```

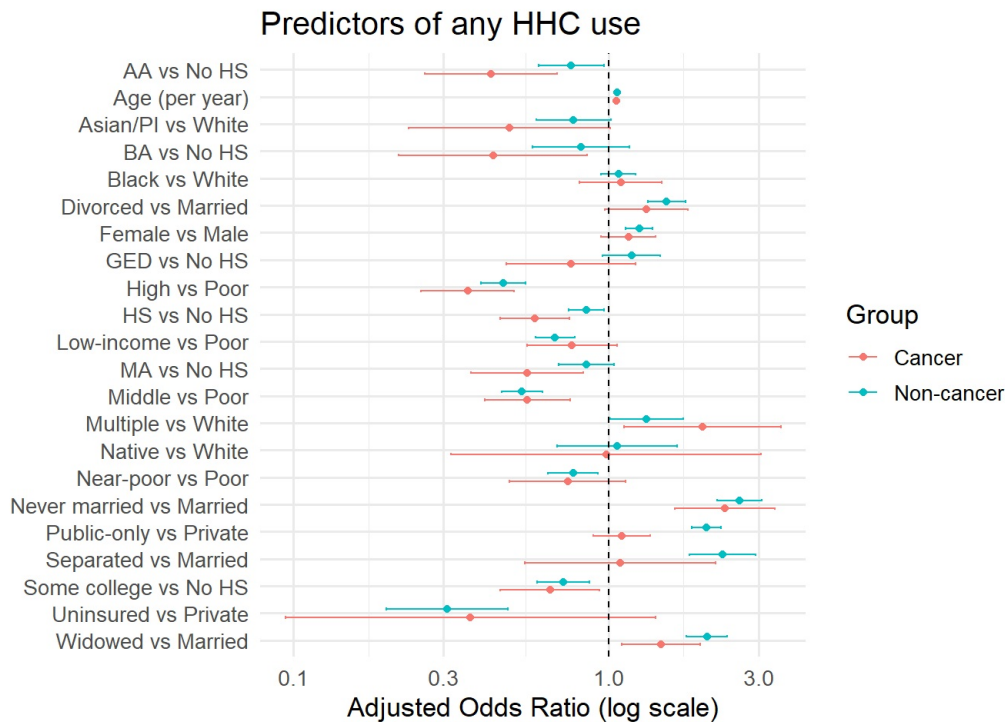


Models — adjusted effects shown as forest plots

```
# Helper to prettify term labels
nice_term <- function(x){
  dplyr::recode(x,
    "AGE" = "Age (per year)",
    "SEXFemale" = "Female vs Male",
    "RACEBlack" = "Black vs White",
    "RACENative" = "Native vs White",
    "RACEAsianPI" = "Asian/PI vs White",
    "RACEMultiple" = "Multiple vs White",
    "INSCOVPublicOnly" = "Public-only vs Private",
    "INSCOVUninsured" = "Uninsured vs Private",
    "POVCATNearPoor" = "Near-poor vs Poor",
    "POVCATLowIncome" = "Low-income vs Poor",
    "POVCATMiddle" = "Middle vs Poor",
    "POVCATHigh" = "High vs Poor",
    "HIDEGGED" = "GED vs No HS",
    "HIDEGHS" = "HS vs No HS",
    "HIDEGSomeCollege" = "Some college vs No HS",
    "HIDEGAA" = "AA vs No HS",
    "HIDEGBA" = "BA vs No HS",
    "HIDEGMA" = "MA vs No HS",
    "MARRYWidowed" = "Widowed vs Married",
    "MARRYDivorced" = "Divorced vs Married",
    "MARRYSeparated" = "Separated vs Married",
    "MARRYNeverMarried" = "Never married vs Married",
    .default = x)
}
```

```
# Utilization (OR)
util_c <- broom::tidy(use_cancer, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)") %>% mutate(group = "Cancer")
util_nc <- broom::tidy(use_noncancer, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)") %>% mutate(group = "Non-cancer")
util_all <- bind_rows(util_c, util_nc) %>% mutate(label = nice_term(term))

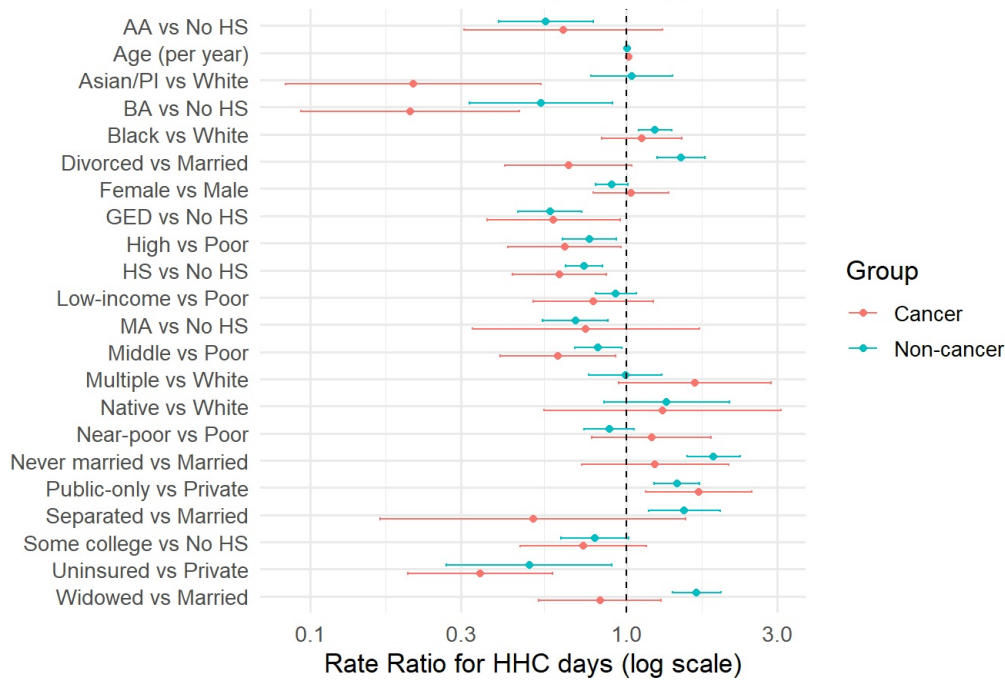
ggplot(util_all, aes(x = estimate, y = forcats::fct_rev(label), color = group))+
  geom_point(position = position_dodge(width = 0.6))+
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2,
    position = position_dodge(width = 0.6))+
  geom_vline(xintercept = 1, linetype = 2) +
  scale_x_log10()+
  labs(x = "Adjusted Odds Ratio (log scale)", y = NULL,
    color = "Group",
    title = "Predictors of any HHC use")+
  theme_minimal(base_size = 13)
```



```
# Intensity (RR for days)
days_c <- broom::tidy(days_cancer, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)") %>% mutate(group = "Cancer")
days_nc <- broom::tidy(days_noncancer, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)") %>% mutate(group = "Non-cancer")
days_all <- bind_rows(days_c, days_nc) %>% mutate(label = nice_term(term))

ggplot(days_all, aes(x = estimate, y = forcats::fct_rev(label), color = group))+
  geom_point(position = position_dodge(width = 0.6))+
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2,
    position = position_dodge(width = 0.6))+
  geom_vline(xintercept = 1, linetype = 2) +
  scale_x_log10()+
  labs(x = "Rate Ratio for HHC days (log scale)", y = NULL,
    color = "Group",
    title = "Predictors of intensity among users")+
  theme_minimal(base_size = 13)
```

Predictors of intensity among users



```
# Costs (percent change)
cost_c <- broom::tidy(cost_cancer, conf.int = TRUE) %>% filter(term != "(Intercept)") %>% mutate(group = "Cancer")
cost_nc <- broom::tidy(cost_noncancer, conf.int = TRUE) %>% filter(term != "(Intercept)") %>% mutate(group = "Non-cancer")

pctify <- function(df){
  df %>% mutate(estimate = 100*(exp(estimate)-1),
               conf.low = 100*(exp(conf.low)-1),
               conf.high = 100*(exp(conf.high)-1))
}

cost_all <- bind_rows(cost_c, cost_nc) %>% pctify() %>% mutate(label = nice_term(term))

ggplot(cost_all, aes(x = estimate, y = forcats::fct_rev(label), color = group))+
  geom_point(position = position_dodge(width = 0.6))+
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2,
                position = position_dodge(width = 0.6))+
  geom_vline(xintercept = 0, linetype = 2) +
  labs(x = "Percent change in total payments (%)", y = NULL, color = "Group",
       title = "Predictors of HHC payments among users")+
  theme_minimal(base_size = 13)
```

Predictors of HHC payments among users

